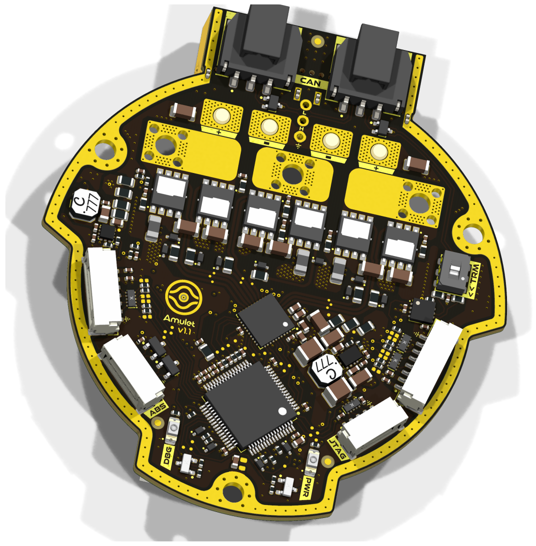
Amulet Motion Controller

2026-08-17

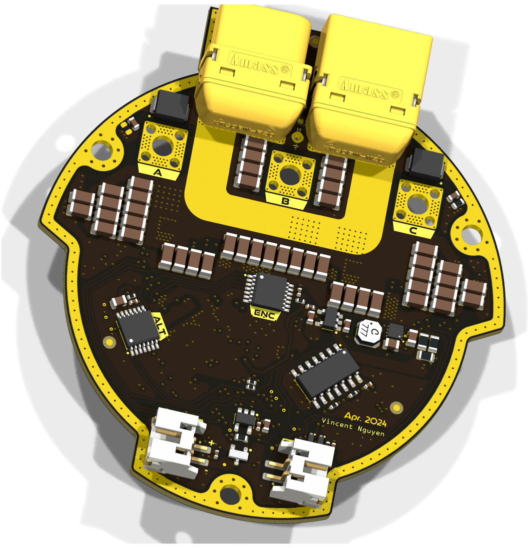
Variant:

Rev 1.1.1+ (Unreleased)

TOP VIEW



BOTTOM VIEW



NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as X

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 17-Aug-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

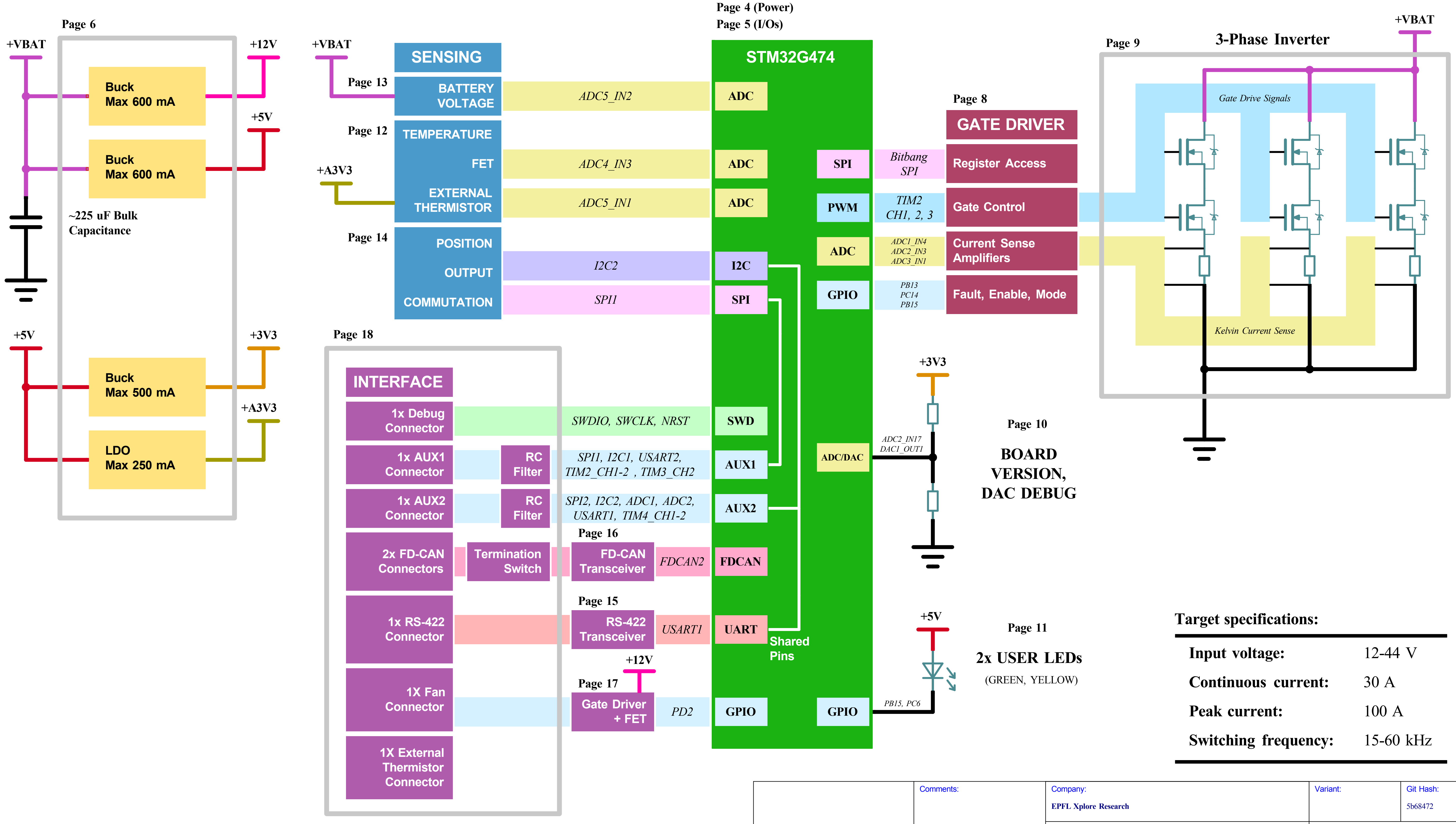
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: **Amulet Motion Controller**		Project Name: **Chienpanzé**	
Sheet Title: Cover Page	File Name: amulet_controller.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 21

[2] Block Diagram

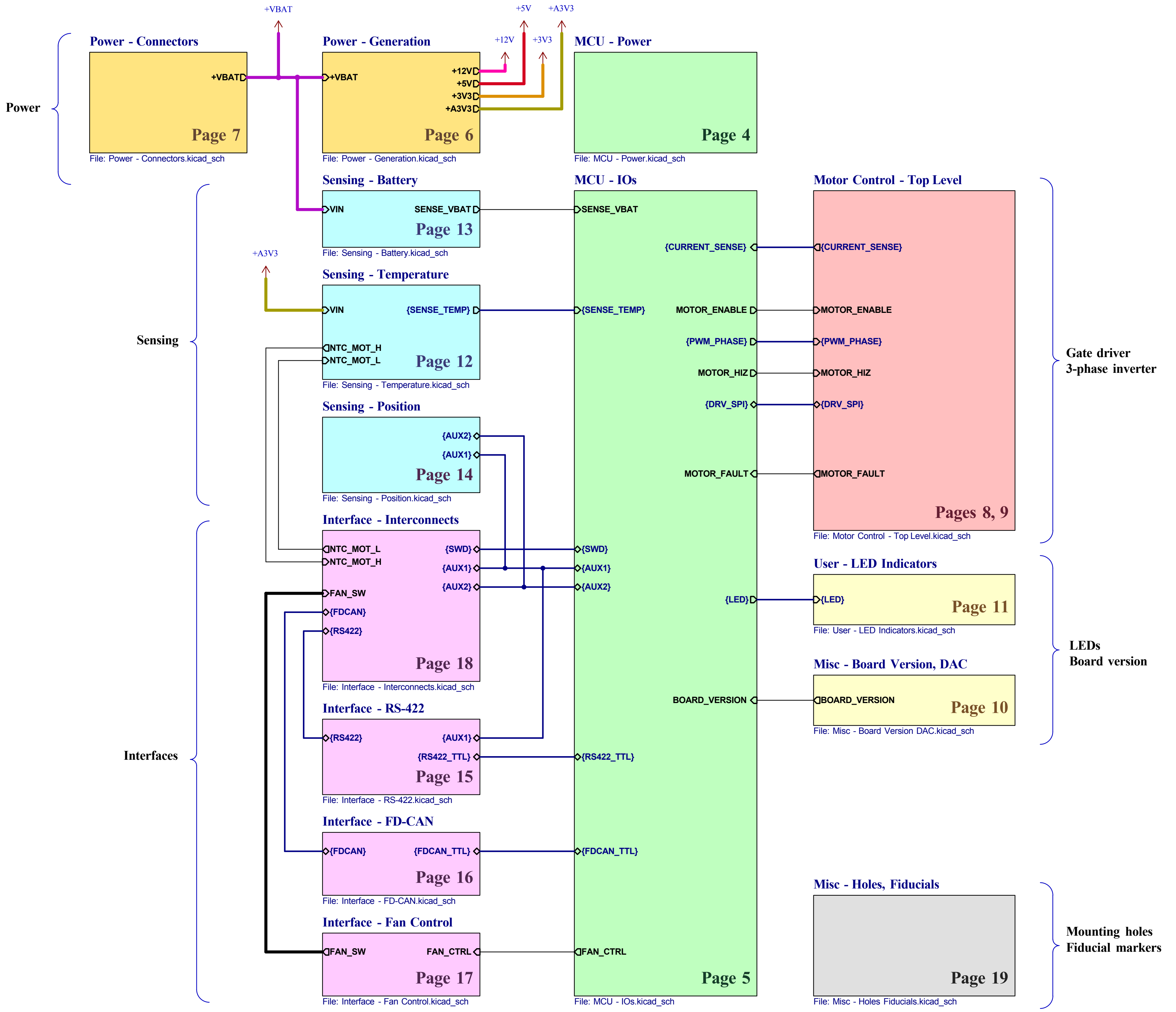


Target specifications:

Input voltage:	12-44 V
Continuous current:	30 A
Peak current:	100 A
Switching frequency:	15-60 kHz

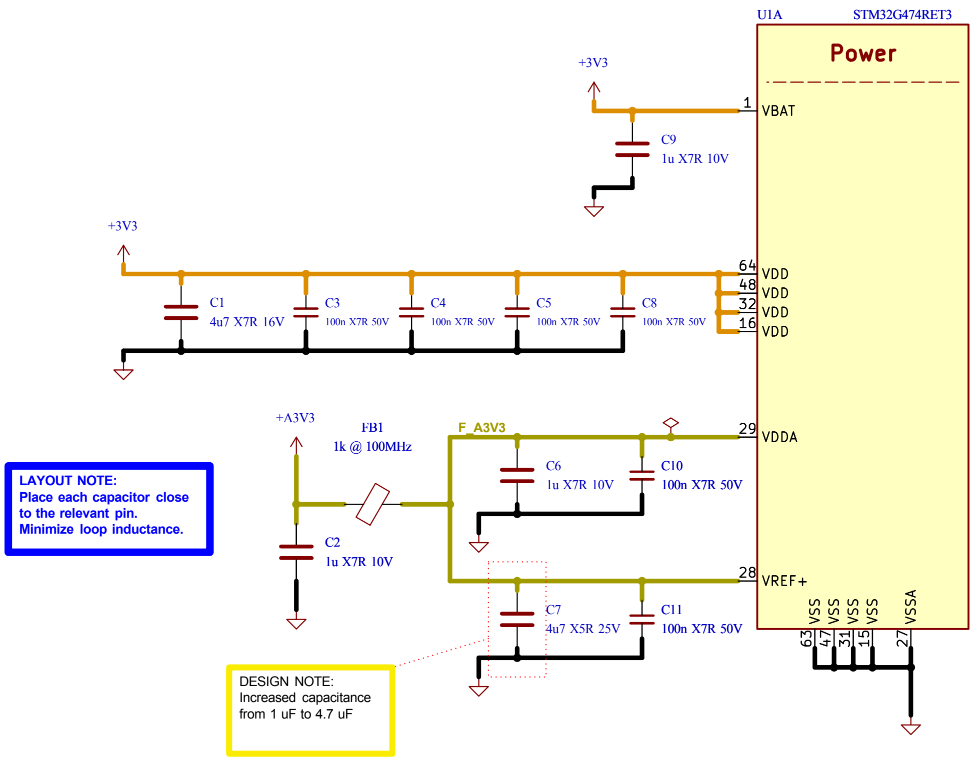
Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 21

[3] Project Architecture



Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: **Amulet Motion Controller**		Project Name: **Chienpanzé**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-22	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 21

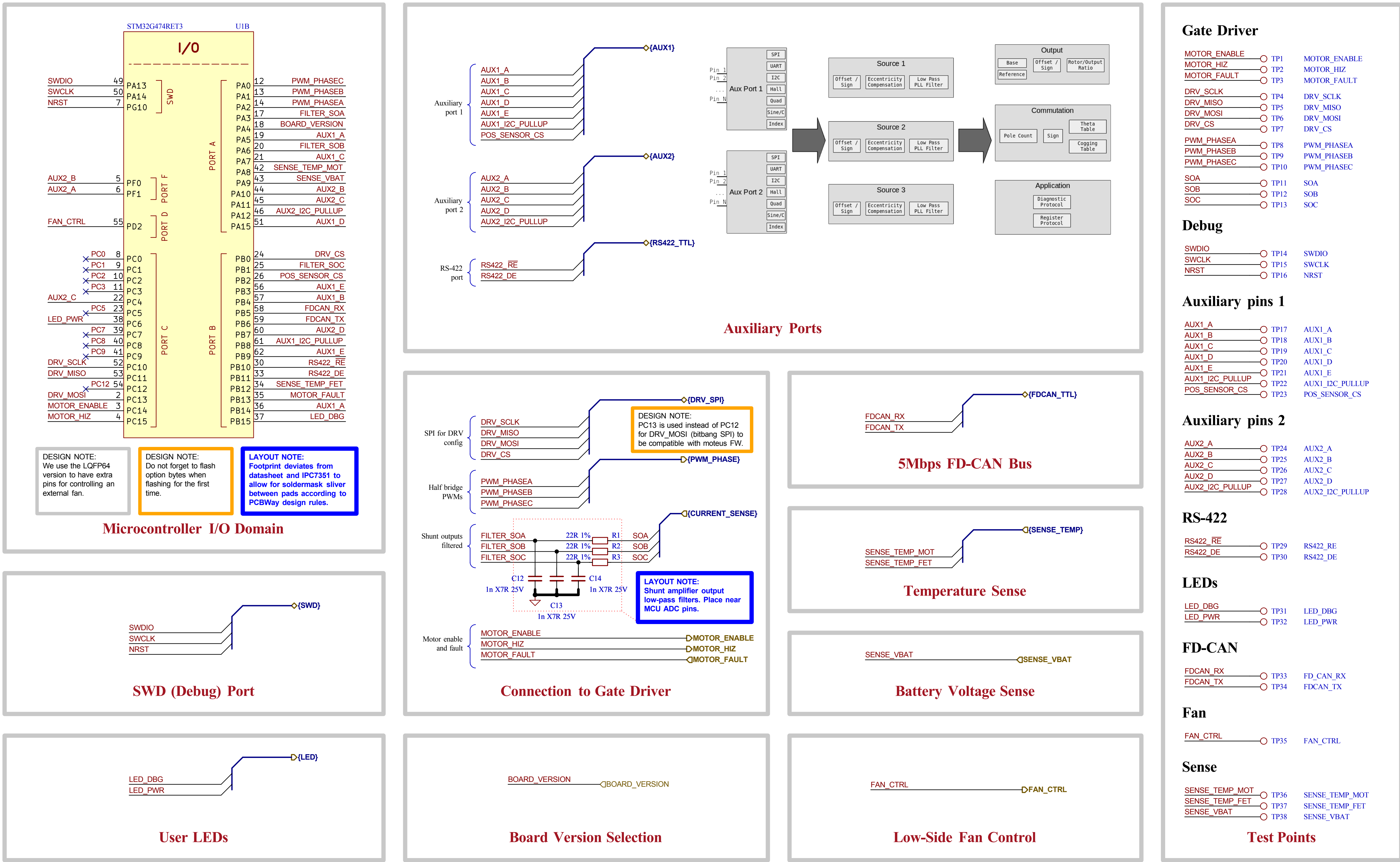
[4] MCU - Power



Microcontroller Power Supply Scheme

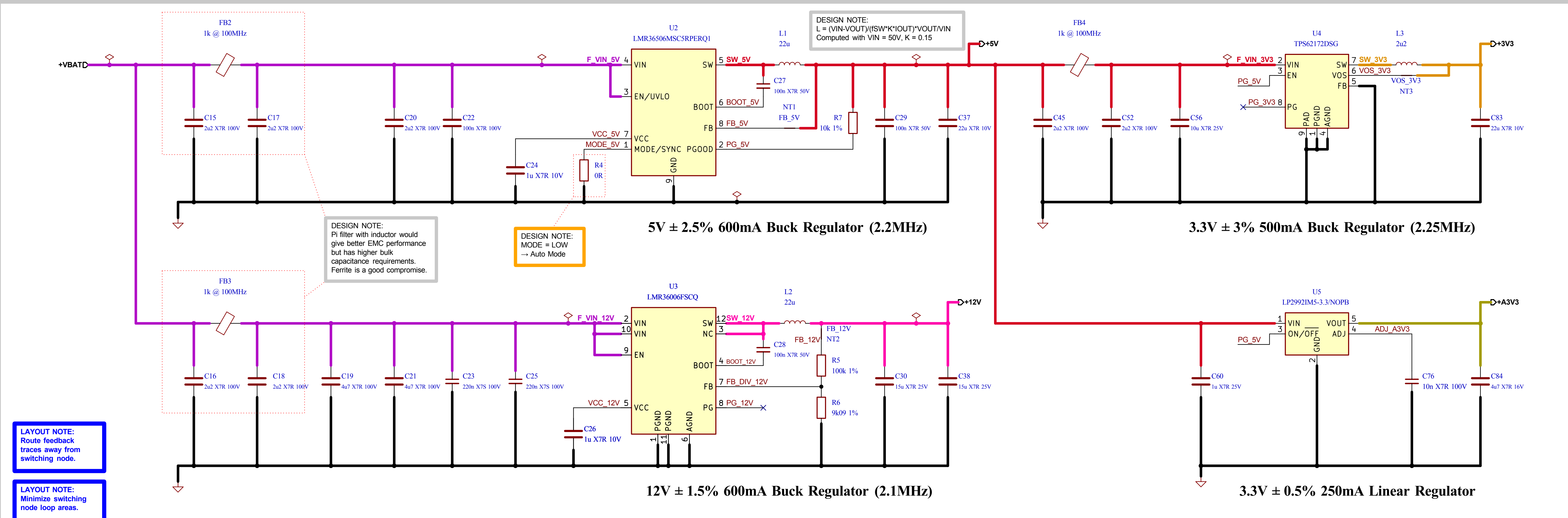
Comments: AN5346 / STM32G474 Datasheet p.81 / J. Pieper ADC investigation	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: MCU - Power	File Name: MCU - Power.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-18	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/MCU - Power/		Reviewer:	Size: A4	Sheet: 4 of 21

[5] MCU - I/Os



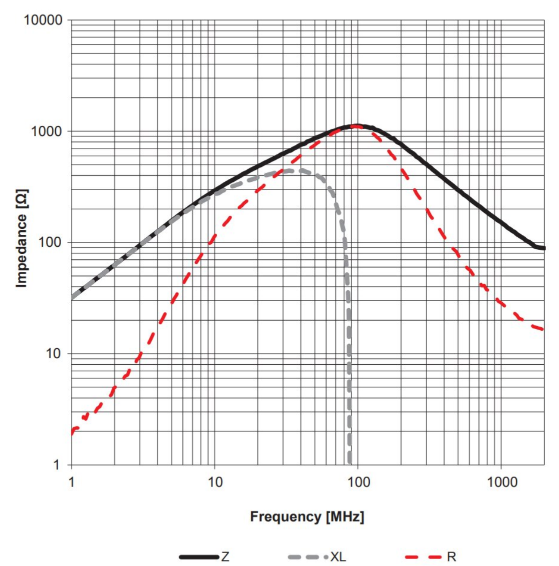
Comments: **References:** Flexible I/O worked examples Flexible I/O source configuration	Company: **EPFL Xplore Research**		Variant:	Git Hash: 5b68472
	Board Name: **Amulet Motion Controller**		Project Name: **Chienpanzé**	
Sheet Title: MCU - I/Os	File Name: MCU - IOs.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-20	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/MCU - IOs/		Reviewer:	Size: **A3**	Sheet: **5** of **21**

[6] Power - Generation



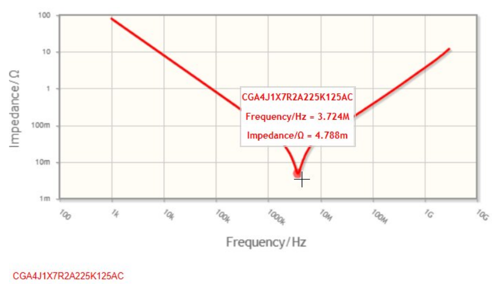
Voltage Regulators

Ferrite bead Impedance vs Frequency curve

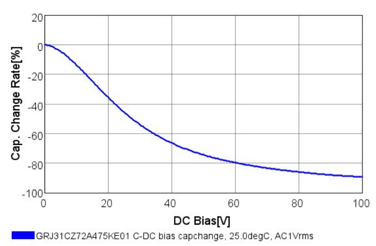


DESIGN NOTE:
XL drops around 45th harmonic.

Pi filter capacitor Impedance vs Frequency curve



Bulk capacitance Derating curve



DESIGN NOTE:
Effective capacitance at 33V is approximately 2.1 uF

Capacitor and Ferrite Bead Parameters

Battery

- +VBAT — TP39 +VBAT
- +VBAT — TP40 +VBAT
- +VBAT — TP41 +VBAT

Filtered Power Rails

- F_VIN_12V — TP42 F_VIN_12V
- F_VIN_5V — TP43 F_VIN_5V
- F_VIN_3V3 — TP44 F_VIN_3V3

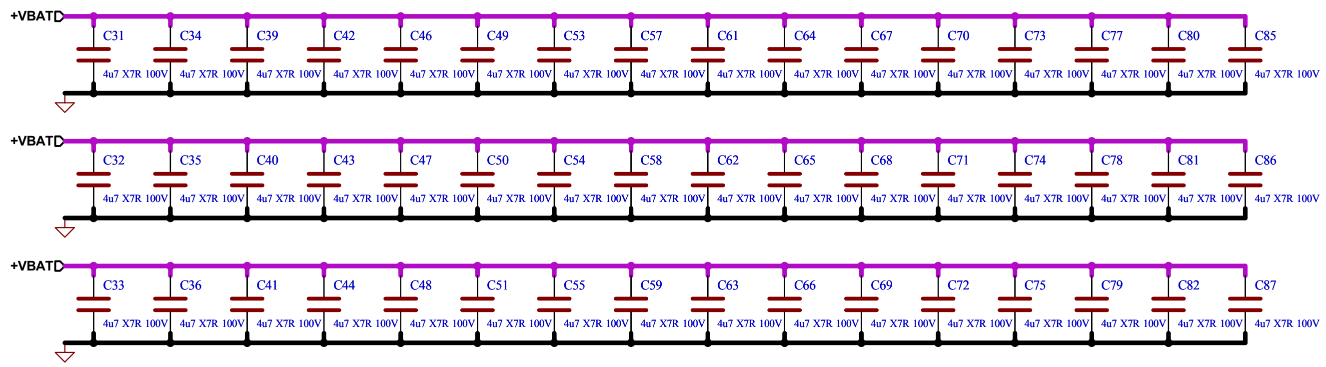
Power Rails

- +12V — TP45 +12V
- +5V — TP46 +5V
- +5V — TP47 +5V
- +3V3 — TP48 +3V3
- +3V3 — TP49 +3V3
- +3V3 — TP50 +3V3
- +A3V3 — TP51 +A3V3

Ground

- GND — TP52 GND
- GND — TP53 GND
- GND — TP54 GND
- GND — TP55 GND
- GND — TP56 GND
- GND — TP57 GND
- GND — TP58 GND

Test Points



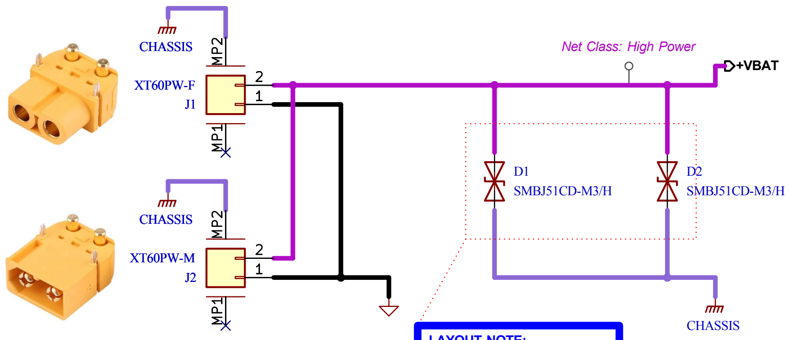
DESIGN NOTE:
An external precharge circuit is advised to charge the bulk capacitors if multiple boards are connected.

DC-Link Bulk Capacitors. 48 x 4.7 uF = 225.6 uF (100.8 uF at 33V)

DESIGN NOTE:
X6S 10uF is marginally better (but worse DC bias derating) at a higher price.

Comments: +12V (600 mA): Fan power +5V (600 mA): Gate Driver, LEDs +3V3 (500 mA): digital 3.3V +3V3A (250 mA): analog 3.3V	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Power - Generation	File Name: Power - Generation.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Power - Generation/		Reviewer:	Size: A3	Sheet: 6 of 21

[7] Power - Connectors



LAYOUT NOTE:
Place TVS diodes near input power connectors.

DESIGN NOTE:
Do not hot-plug when powered!

DESIGN NOTE:
Do not exceed absolute maximum ratings! Maximum input voltage is 44V (10S).

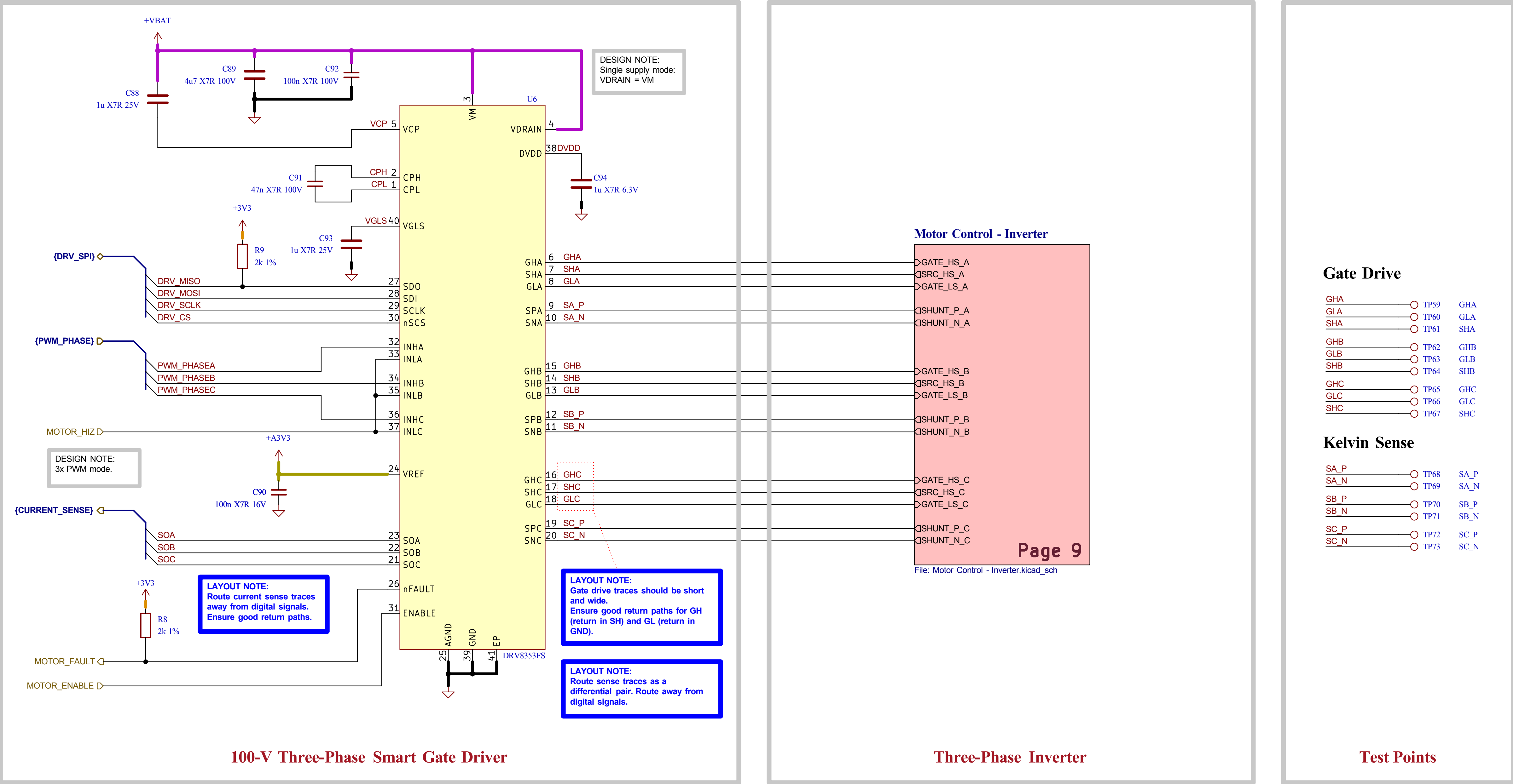
LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1

DESIGN NOTE:
It's best to power the controller through the male connector to always have a female connector at the end of the chain.

Power Connectors, TVS Diodes

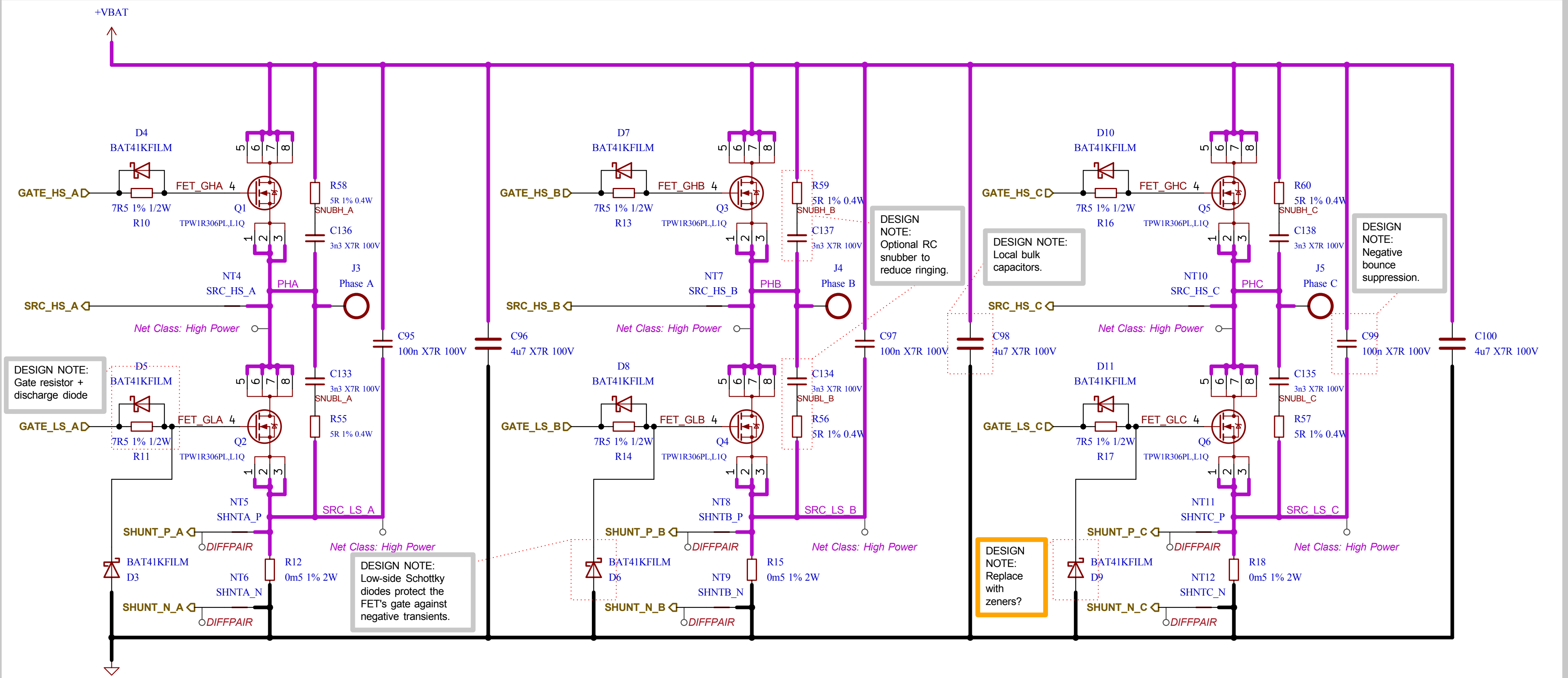
Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Power - Connectors	File Name: Power - Connectors.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Power - Connectors/		Reviewer:	Size: A4	Sheet: 7 of 21

[8] Motor Control - Top Level



Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Motor Control - Top Level	File Name: Motor Control - Top Level.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-20	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Top Level/		Reviewer:	Size: A3	Sheet: 8 of 21

[9] Motor Control - Inverter



Three-Phase Inverter

LAYOUT NOTE:
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

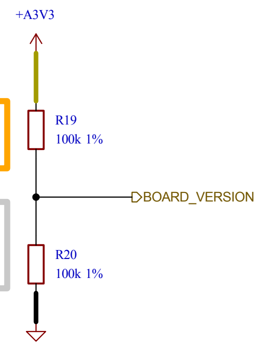
DESIGN NOTE:
A gate drive current that is too large can damage the FETs!

Comments: System Design Considerations for High-Power Motor Driver Applications; Best Practices for Board Layout of Motor Drivers; Proper RC Snubber Design for Motor Drivers	Company: EPFL Xplore Research			Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller			Project Name: Chienpanzé	
Sheet Title: Motor Control - Inverter	File Name: Motor Control - Inverter.kicad_sch	Designer: Vincent Nguyen		Date: 2024-01-25	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Top Level/Motor Control - Inverter/		Reviewer:		Size: A4	Sheet: 9 of 21

[10] Misc - Board Version, DAC

DESIGN NOTE:
Necessary to be compatible with moteus FW.

DESIGN NOTE:
Pin is also used at compile time to output analog debug data (DAC).

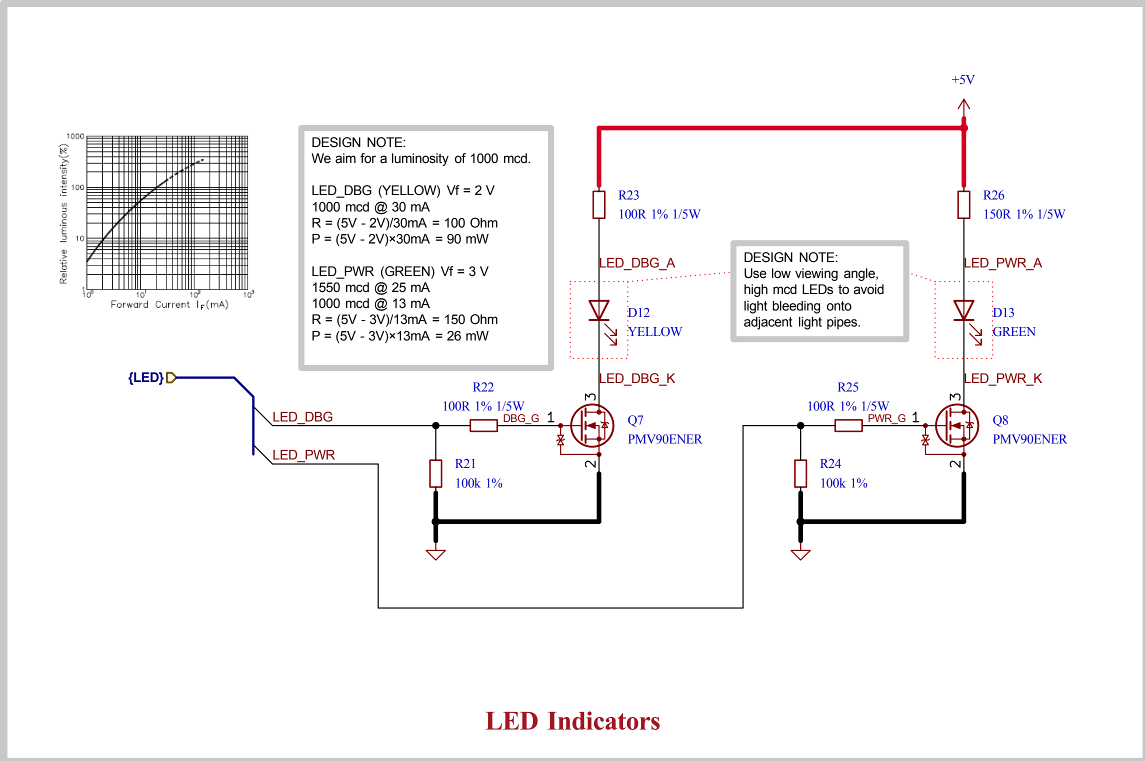


		TOP	BOTTOM
moteus	r1.1, r1.2	×	100k
	r1.3	100k	×
Amulet	v1.0-1.1	100k	100k

Board Version Resistor Divider

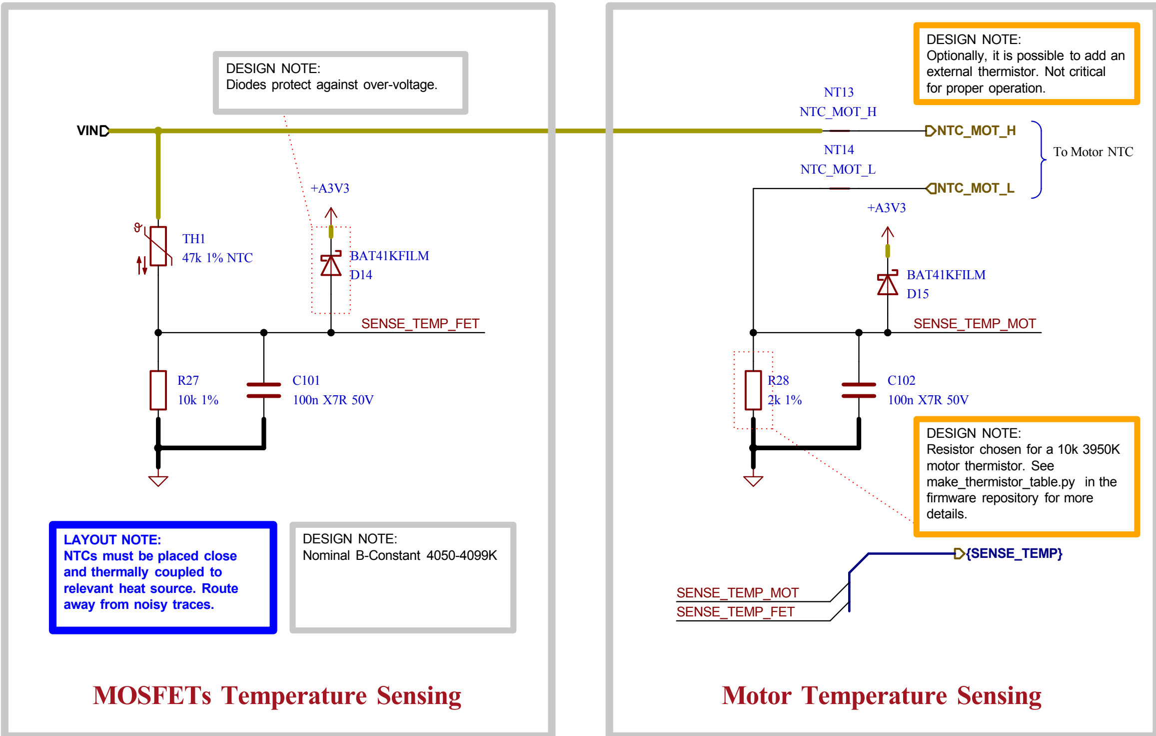
Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: **Amulet Motion Controller**		Project Name: **Chienpanzé**	
Sheet Title: Misc - Board Version, DAC	File Name: Misc - Board Version DAC.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Misc - Board Version, DAC/		Reviewer:	Size: **A4**	Sheet: **10** of **21**

[11] User - LED Indicators



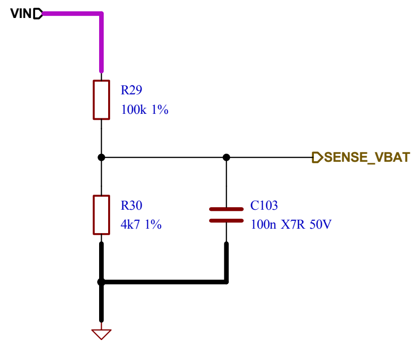
Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: User - LED Indicators	File Name: User - LED Indicators.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-19	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/User - LED Indicators/		Reviewer:	Size: A4	Sheet: 11 of 21

[12] Sensing - Temperature



Comments:	Company: EPFL Xplore Research	Variant:	Git Hash: 5b68472
	Board Name: **Amulet Motion Controller**	Project Name: **Chienpanzé**	
Sheet Title: Sensing - Temperature	File Name: Sensing - Temperature.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13 / Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Sensing - Temperature/		Reviewer:	Size: A4 / Sheet: 12 of 21

[13] Sensing - Battery



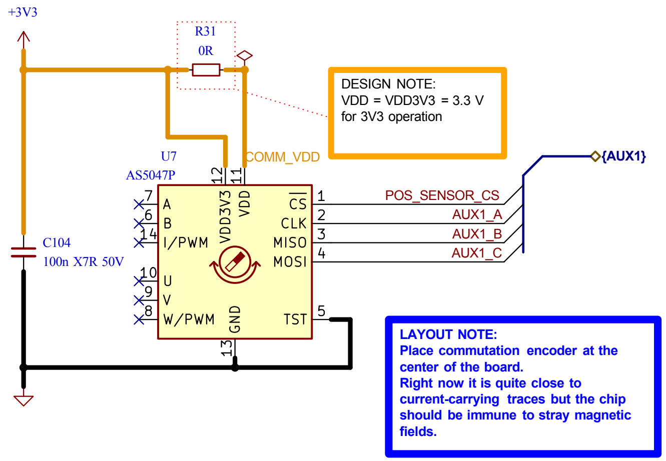
DESIGN NOTE:
1/23 temperature stable voltage divider. Use low ppm resistors.

12S: max voltage: 2.19 V
8S: max voltage: 1.95 V
6S: max voltage: 1.46 V

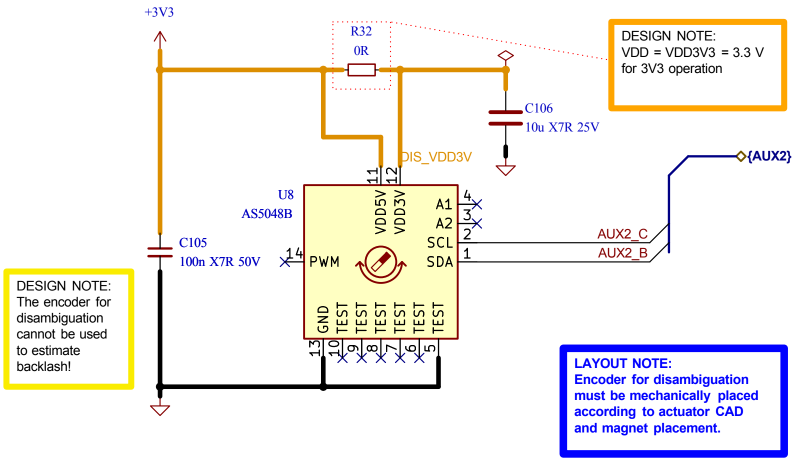
Battery Voltage Sensing

Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: **Amulet Motion Controller**		Project Name: **Chienpanzé**	
Sheet Title: Sensing - Battery	File Name: Sensing - Battery.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-14	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer:	Size: A4	Sheet: 13 of 21

[14] Sensing - Position



14-Bit On-Axis Magnetic Rotary Position Sensor (commutation)

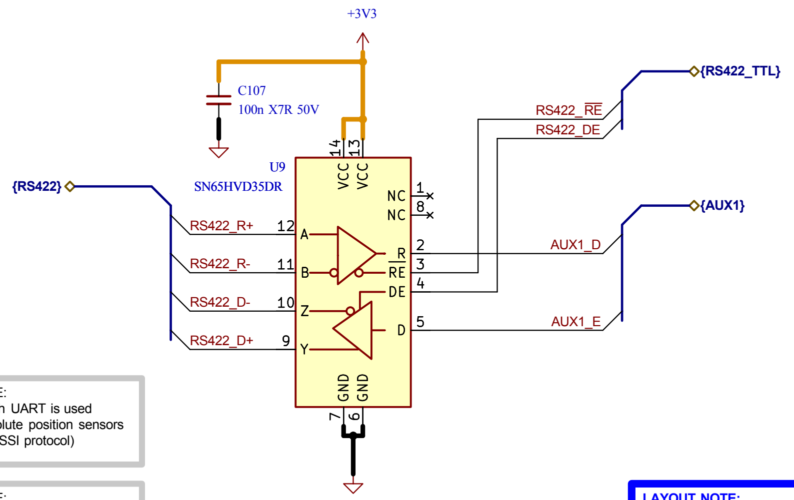


14-Bit On-Axis Magnetic Rotary Position Sensor (disambiguation)

DESIGN NOTE:
AS5047P senses magnet mounted on planetary sun gear, for commutation.
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.

Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: **Amulet Motion Controller**		Project Name: **Chienpanzé**	
Sheet Title: Sensing - Position	File Name: Sensing - Position.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-14	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Sensing - Position/		Reviewer:	Size: A4	Sheet: 14 of 21

[15] Interface - RS-422



DESIGN NOTE:
RS-422 through UART is used mostly for absolute position sensors (e.g. Aksim-2, SSI protocol)

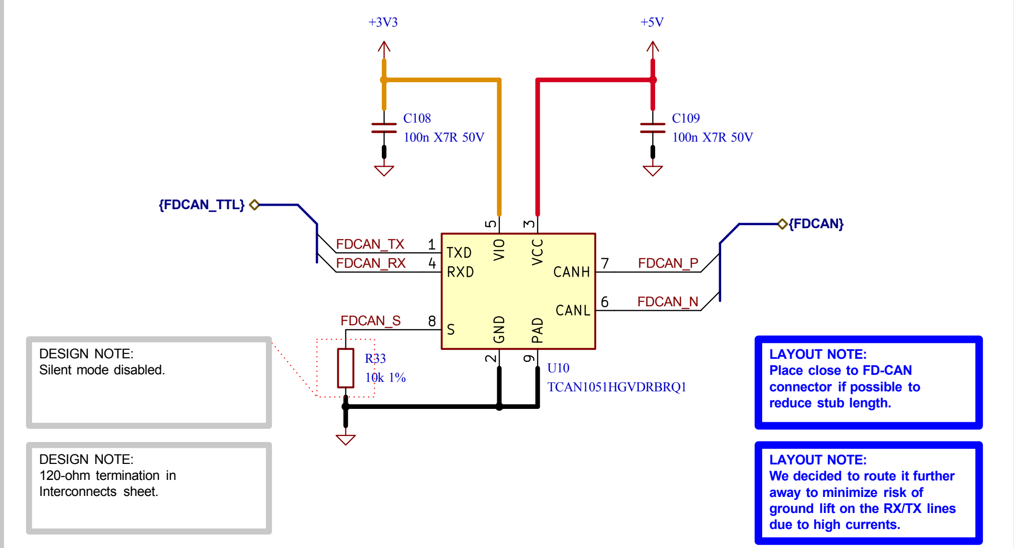
DESIGN NOTE:
100-ohm termination in Interconnects sheet.

LAYOUT NOTE:
Place close to RS-422 connector.

3.3-V Full-Duplex RS-422 Interface

Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Interface - RS-422	File Name: Interface - RS-422.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Interface - RS-422/		Reviewer:	Size: A4	Sheet: 15 of 21

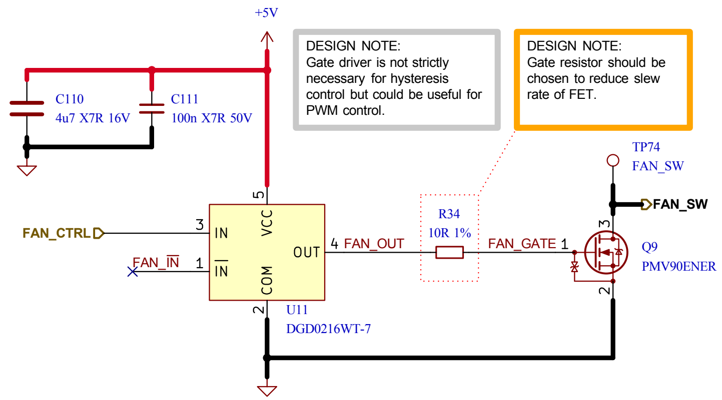
[16] Interface - FD-CAN



Fault Protected FD-CAN Transceiver

Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: **Amulet Motion Controller**		Project Name: **Chienpanzé**	
Sheet Title: Interface - FD-CAN	File Name: Interface - FD-CAN.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-15	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: **A4**	Sheet: **16** of **21**

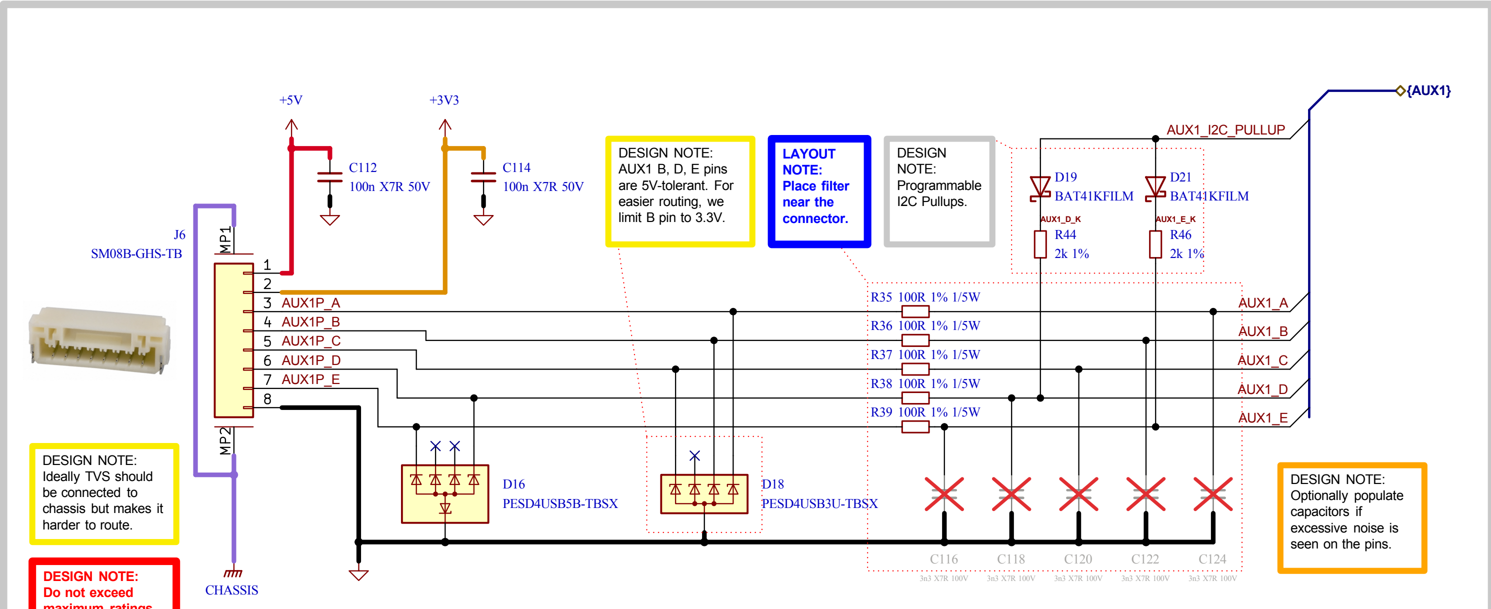
[17] Interface - Fan Control



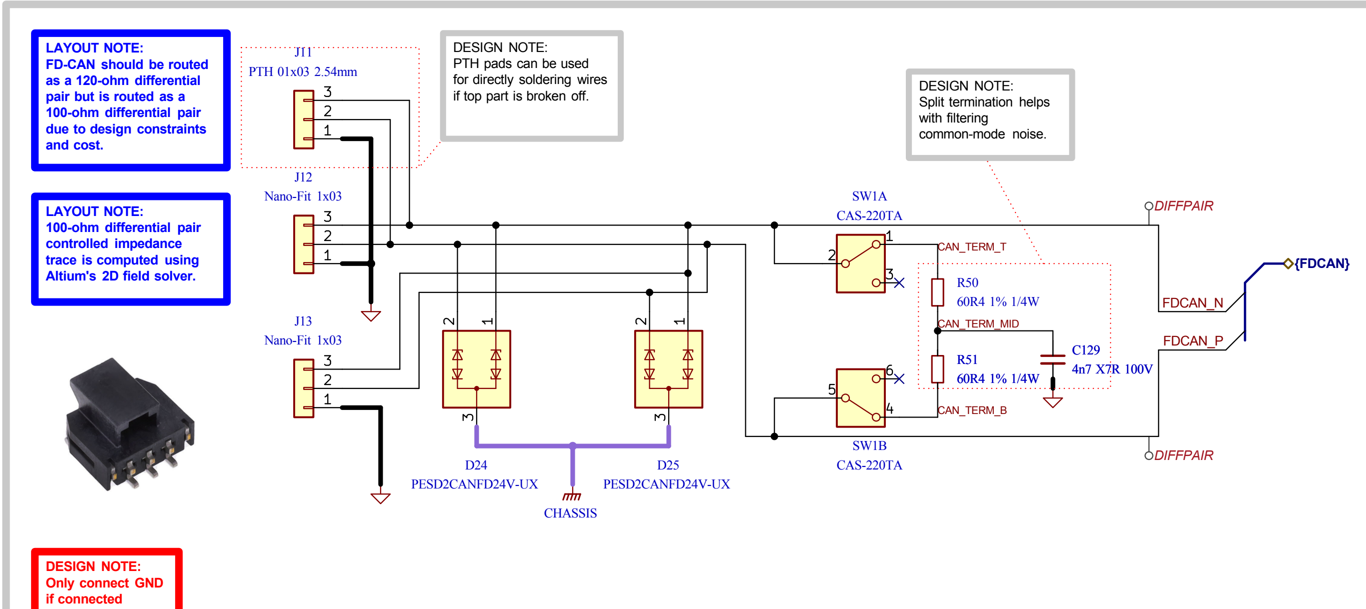
Low-side Fan Switch and Gate Driver

Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Interface - Fan Control	File Name: Interface - Fan Control.kicad_sch	Designer: Vincent Nguyen	Date: 2023-11-19	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Interface - Fan Control/		Reviewer:	Size: A4	Sheet: 17 of 21

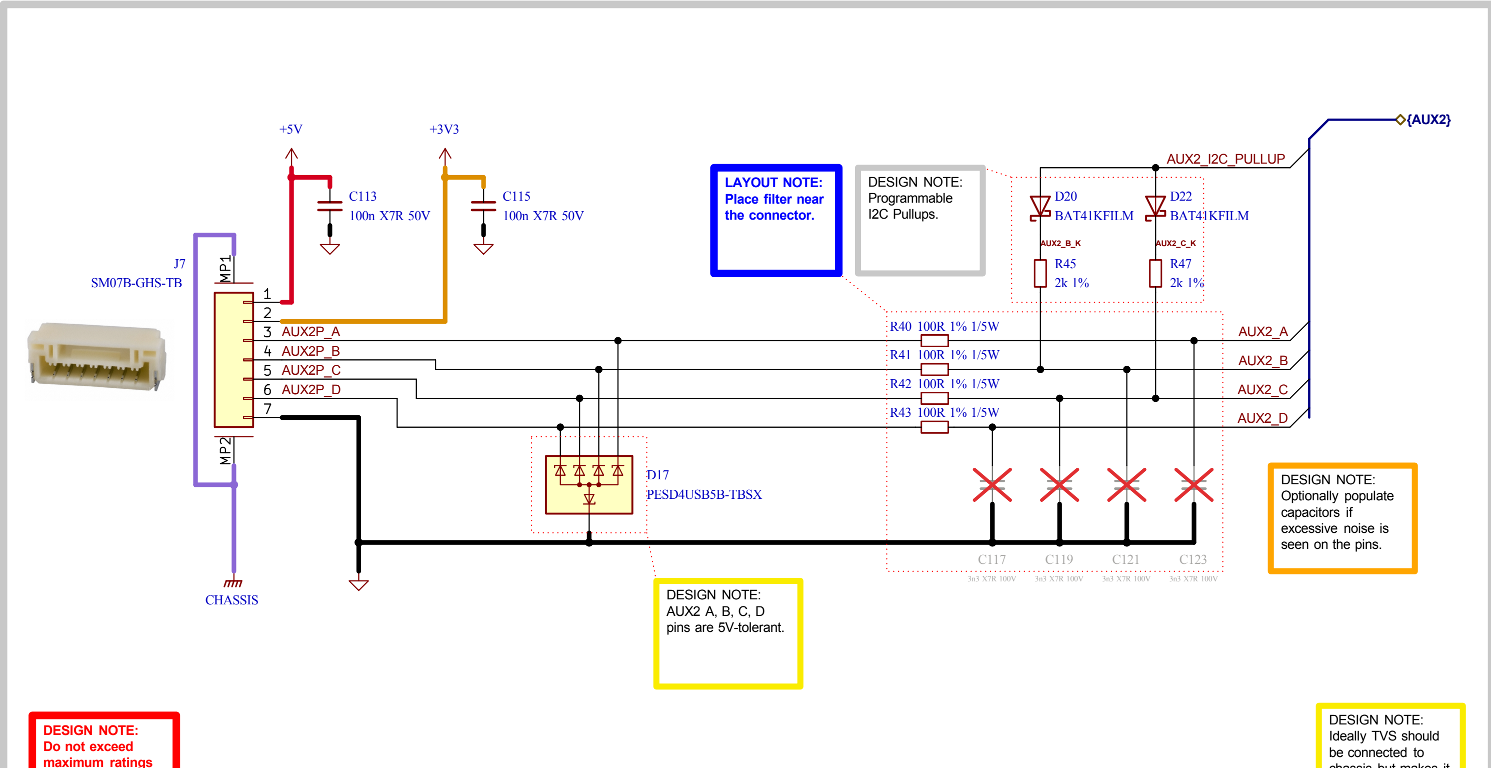
[18] Interface - Interconnects



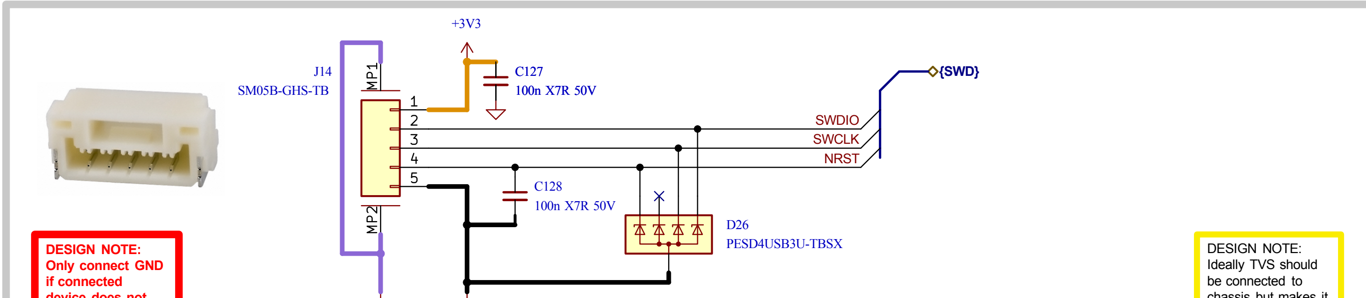
Auxiliary Connector AUX1



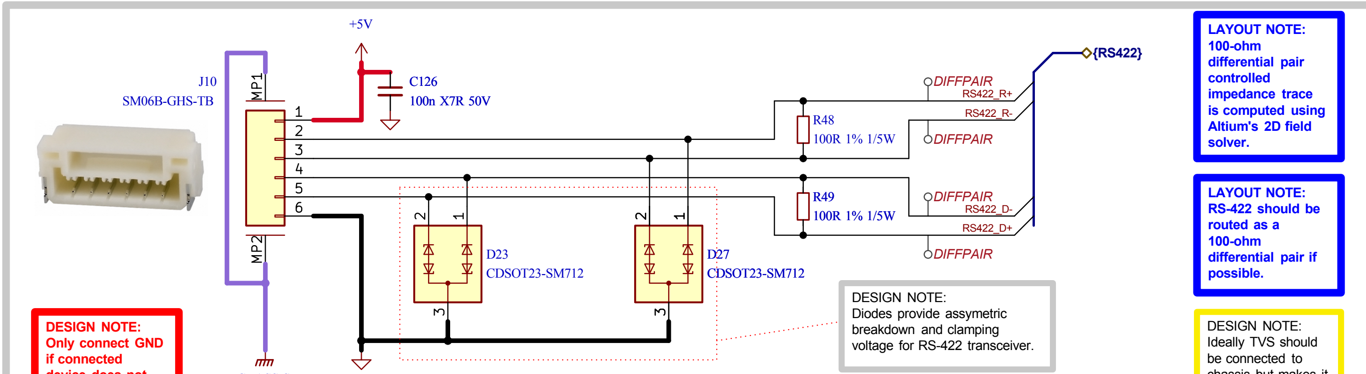
FD-CAN Connector and Termination



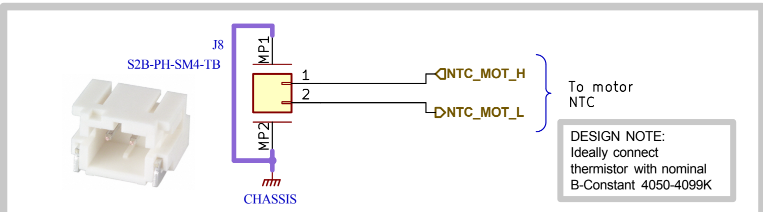
Auxiliary Connector AUX2



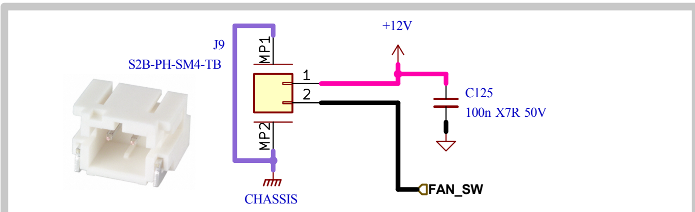
SWD Debug Connector



RS-422 Connector



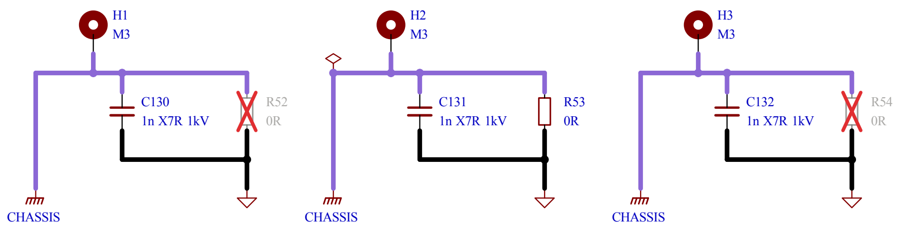
Motor Temperature Sensing Thermistor Connection



12V Fan Connector

Comments:	Company:		Variant:	Git Hash:
Reference: Flexible I/O worked examples	EPFL Xplore Research			5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Interface - Interconnects	File Name: Interface - Interconnects.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer:	Size: A3	Sheet: 18 of 21

[19] Misc - Holes, Fiducials



DESIGN NOTE:
Optional RC filter for chassis grounding. Capacitors offer multi-point low-impedance connections at high-frequencies.

DESIGN NOTE:
0-Ohm resistor provides low-impedance bond between chassis and signal GND. Be aware of ground loops through cable shield.

Mounting Holes and Chassis Connection

Front

Back

Fiducial Markers

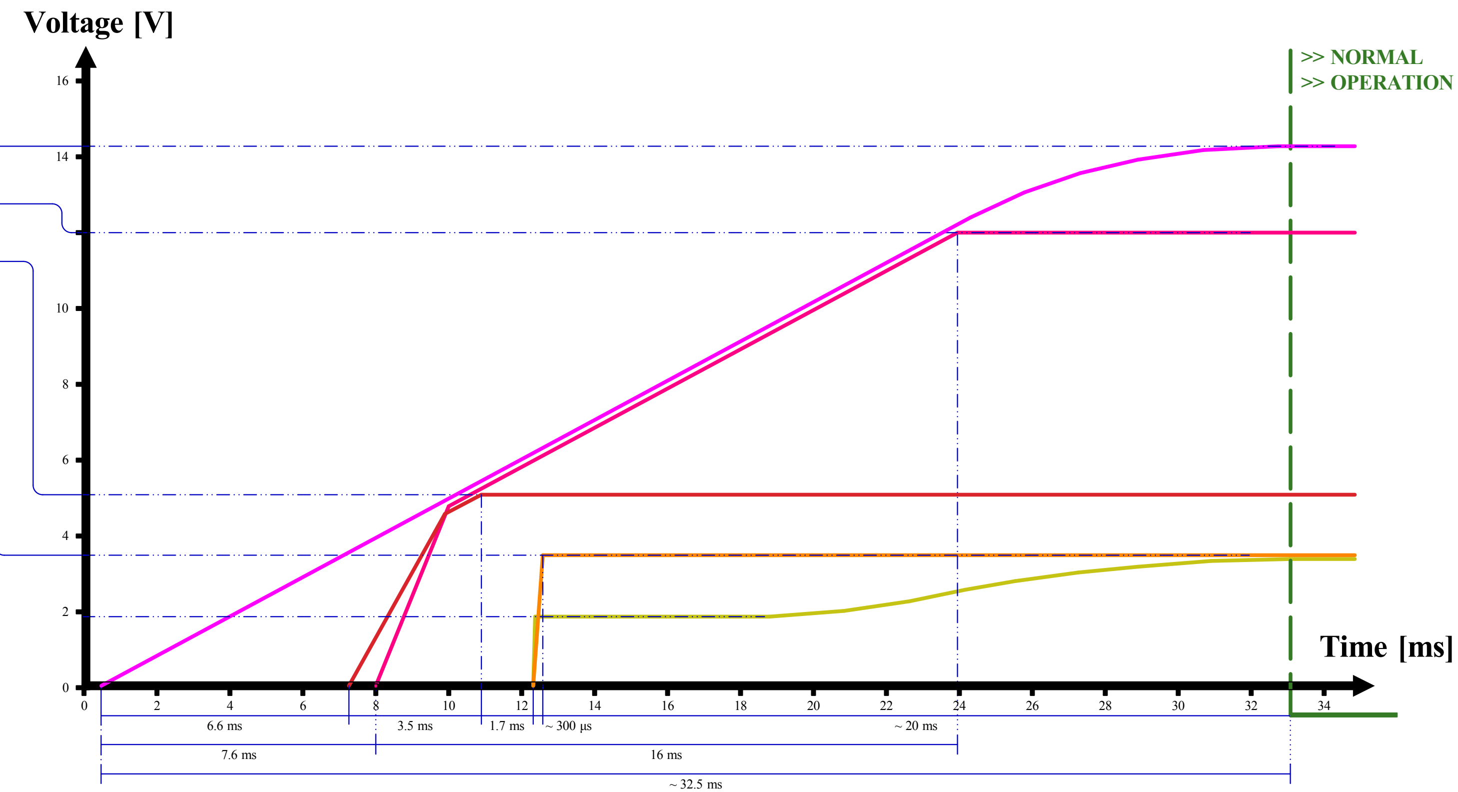
Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Misc - Holes, Fiducials	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Vincent Nguyen	Date: 2023-10-22	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 19 of 21

[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V ± 1.5%
+5V	LMR36506	5V ± 1.5%
+3V3	TPS62172	3.3V ± 3%
+A3V3	LP2992	3.3V ± 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Vincent Nguyen	Date: 2024-03-12	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 20 of 21

[21] Revision History

Version 1.0.0 - 2023-12-12

Added

- TVS protection and termination switch to FD-CAN.
- Low-side switched 12V 600 mA source for external fan.
- LDO for analog supply.
- One TVS diode per power connector.
- Second on-board I2C magnetic encoder for disambiguation.
- ESD protection to all interfaces.
- Over-voltage protection on thermistor ADC inputs.
- Pi filters to inputs of buck regulators and MCU analog supply.
- Decoupling caps next to power pins of connectors.

Changed

- CPH-CPL capacitor to 47 nF (gate driver).
- FD-CAN transceiver IC.
- FETs for top cooled variant.
- Input power TVS diode to bidirectional.
- Moved SOx low-pass filter to MCU section.
- PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing.
- RS422 pinout on connector.
- Buck regulators to optimize for low noise.

Version 1.0.1 - 2024-01-25

Added

- Controller target specifications.
- Credits to moteus on cover page.
- Optional RC-Snubber to power stage.

Fixed

- Chassis guard ring to go around the board.
- CAN and power TVS diodes now go to chassis.
- Clearance between nets to respect IEC60664-1 where possible.
- Comment on precharge.
- Power TVS diode reference designator from "U" to "D".

Changed

- 5V 300 mA buck converter with 600 mA version.
- Chassis-GND capacitor by 1nF 1kV.

Version 1.0.2 - 2024-03-12

Changed

- Power sequencing graph according to experimental data.

Version 1.1.0 - 2024-04-13

Added

- RC snubber passive values.

Fixed

- More vias for VBUS and LMR36006 GND pads.
- Board version voltage reference from +3V3 to +A3V3.

Changed

- Motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.

Version 1.1.1 - 2025-01-18

Fixed

- Replace non-stocked components.

Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Vincent Nguyen	Date: 2024-01-03	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 21 of 21